

CSE 400: Fundamentals of Probability in
Computing
Lecture 20: Linear Transformations and
Expectations of Multiple RVs

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1 Random Vector Representation

1.1 Vector Form of Multiple Random Variables

For random variables $X_1, X_2, \dots, X_N$, we represent them as a vector:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Definition: X is called a random vector.

2 Mean Vector

2.1 Expectation of a Random Vector

Definition: The mean vector of X is:

$$\mu_X = E[X]$$

Example structure:

$$\mu_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

3 Correlation Matrix

3.1 Correlation Between Multiple RVs

Definition:

$$R_{XX} = E[XX^T]$$

Element-wise:

$$R_{XX}(i, j) = E[X_i X_j], \quad i, j \leq N$$

Matrix form:

$$R_{XX} = \begin{bmatrix} E[X_1 X_1] & E[X_1 X_2] & \cdots & E[X_1 X_N] \\ E[X_2 X_1] & E[X_2 X_2] & \cdots & E[X_2 X_N] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_N X_1] & E[X_N X_2] & \cdots & E[X_N X_N] \end{bmatrix}$$

Property: R_{XX} is symmetric.

4 Covariance Matrix

4.1 Definition

$$C_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$$

Matrix structure:

$$C_{XX} = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Property:

$$\text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i)$$

5 Linear Transformation of Random Vectors

5.1 Linear Model

$$Y = AX + b$$

For N -dimensional vectors:

$$Y_i = \sum_{j=1}^N a_{ij} X_j + b_i$$

6 Mean of Transformed Vector

6.1 Definition

$$\mu_Y = E[Y]$$

6.2 Derivation

$$\begin{aligned}\mu_Y &= E[AX + b] \\ &= E[AX] + E[b]\end{aligned}$$

Result:

$$\mu_Y = A\mu_X + b$$

7 Correlation Matrix Transformation

7.1 Definition

$$R_{YY} = E[YY^T] = E[(AX + b)(AX + b)^T]$$

7.2 Expansion

$$R_{YY} = E[AXX^T A^T + AXb^T + bX^T A^T + bb^T]$$

7.3 Applying Expectation

$$R_{YY} = AE[XX^T]A^T + AE[X]b^T + bE[X]^T A^T + bb^T$$

Final Result:

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

8 Covariance Matrix Transformation

8.1 Key Observation

$$Y - \mu_Y = (AX + b) - (A\mu_X + b) = A(X - \mu_X)$$

8.2 Derivation

$$\begin{aligned} C_{YY} &= E[(Y - \mu_Y)(Y - \mu_Y)^T] \\ &= E[A(X - \mu_X)(X - \mu_X)^T A^T] \end{aligned}$$

Simplified Result:

$$C_{YY} = AC_{XX}A^T$$

Note: Shift b does not affect covariance.

9 Summary

If $Y = AX + b$:

$$\begin{aligned} \mu_Y &= A\mu_X + b \\ C_{YY} &= AC_{XX}A^T \\ R_{YY} &= AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T \end{aligned}$$

10 Example: Numerical Linear Transformation

Given:

$$\mu_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$R_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}$$

10.1 (a) Mean Vector

$$\mu_Y = A\mu_X$$

$$= \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

10.2 (b) Correlation Matrix

Step 1:

$$AR_{XX} = \begin{bmatrix} 10 & -1 & -7 \\ 1 & -7 & 7 \\ -11 & 8 & 0 \end{bmatrix}$$

Step 2:

$$R_{YY} = (AR_{XX})A^T = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}$$

10.3 (c) Covariance Matrix

$$C_{YY} = R_{YY} - \mu_Y \mu_Y^T$$

$$\mu_Y \mu_Y^T = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \begin{bmatrix} 1 & 2 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$$

$$C_{YY} = \begin{bmatrix} 17 & -6 & -9 \\ -6 & 14 & -8 \\ -9 & -8 & 19 \end{bmatrix} - \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}$$

11 Gaussian Distributions

11.1 Univariate Gaussian (1D)

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

11.2 Joint Gaussian (2D)

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}} \exp\left(-\frac{1}{2(1-\rho_{XY}^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho_{XY} \left(\frac{x-\mu_X}{\sigma_X}\right) \left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]\right)$$

12 Multivariate Gaussian Distribution

$$X = [X_1, X_2, \dots, X_N]^T$$

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(x-\mu_X)^T C_{XX}^{-1} (x-\mu_X)\right)$$

13 Example: 2D Joint Gaussian

13.1 Determinant

$$\det(C_{XX}) = \sigma_1^2 \sigma_2^2 (1 - \rho^2)$$

13.2 Inverse

$$C_{XX}^{-1} = \frac{1}{\det(C_{XX})} \begin{bmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{bmatrix}$$

13.3 Quadratic Form

$$Q(x) = (x - \mu_X)^T C_{XX}^{-1} (x - \mu_X)$$

$$Q(x) = \frac{1}{1 - \rho^2} \left[\left(\frac{x_1 - \mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{x_1 - \mu_1}{\sigma_1} \right) \left(\frac{x_2 - \mu_2}{\sigma_2} \right) + \left(\frac{x_2 - \mu_2}{\sigma_2} \right)^2 \right]$$

14 Uncorrelated Gaussian Variables

$$f_X(x) = \prod_{n=1}^N \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp \left(-\frac{(x_n - \mu_n)^2}{2\sigma_n^2} \right)$$

Condition:

$$\text{Cov}(X_i, X_j) = 0, \quad i \neq j$$

15 Homework

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$